Graphic Organizer – Adding, Subtracting and Multiplying

Adapted from an activity via http://www.ilovemath.org

Guidelines for Adding, Subtracting, and Multiplying Complex Numbers

Adding

Parentheses do not change the problem. Combine like terms (real with real and imaginary with imaginary) – by, combining coefficients.

- Examples -

(7 + 6*i*) + (8 - 3*i*) =

(-5 - 2*i*) + (-8*i*) =

Subtracting

All signs for each term must be switched in the set of parentheses that follow the subtraction sign. Then follow the rules for adding complex numbers.

- Examples -

(12 + *i*) – (-4 + 5*i*) =

(2) – (8 – 7*i*) =

Multiplying

When multiplying two complex numbers, you are just using the distributive property multiple times. Simplify the resulting expression. Remember $i^2 = -1$

- Examples -

(3 + 5*i*) (6 – 9*i*) =

(6 – 3*i*) (6 + 3*i*) =

After practicing the operations above we will move on to figuring out how we can **divide** Complex Numbers.

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Division of Complex Numbers

Write the following in the form $a + ib$

1.	$\frac{9-6i}{3}$	
2.	$\frac{1}{i}$	
3.	$\frac{7-4i}{1-2i}$	
4.	$\frac{3+i}{3-i}$	
5.	$\frac{2-4i}{-i}$	
6.	$\frac{1}{5-4i}$	
7.	$(5-4i)\left(\frac{5}{41}+\frac{4i}{41}\right)$	
8.	Find the multiplicative inverse of 3 - 2i	
9.	Calculate the quotient of $\frac{1+7i}{1+2i}$ Plot 1 + 7i, 1 + 2i and their quotient on an Argand Diagram Calculate | 1 + 7i |, | 1 + 2i |,|quotient| and investigate if $\frac{\lvert 1+7i\rvert}{\lvert 1+2i\rvert}$ = |quotient|. Calculate the angles that 1 + 7i and 1 + 2i make with the Real axis and investigate if the subtraction of these angles is equal to the angle that the quotient makes with the Real axis.	
10.	$z_1 = a + bi$, $z_2 = c + di$, Find: $\frac{z_1}{z_2} =$ $\frac{z_2}{z_1} =$	

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